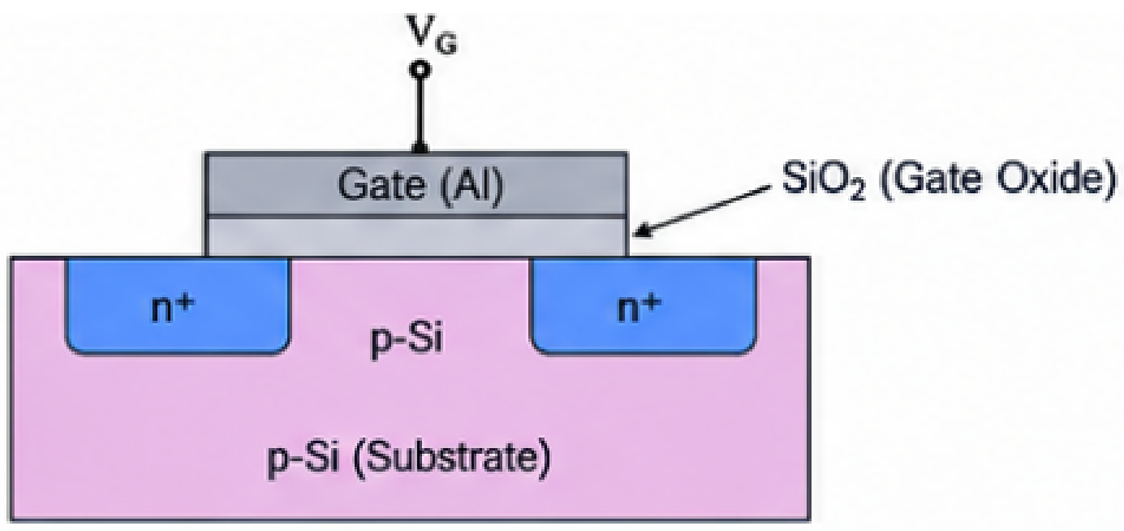
EXPERIMENT 3

Simulation-Based Analysis of MOS Structure and Gate-Oxide Formation Using NanoHUB

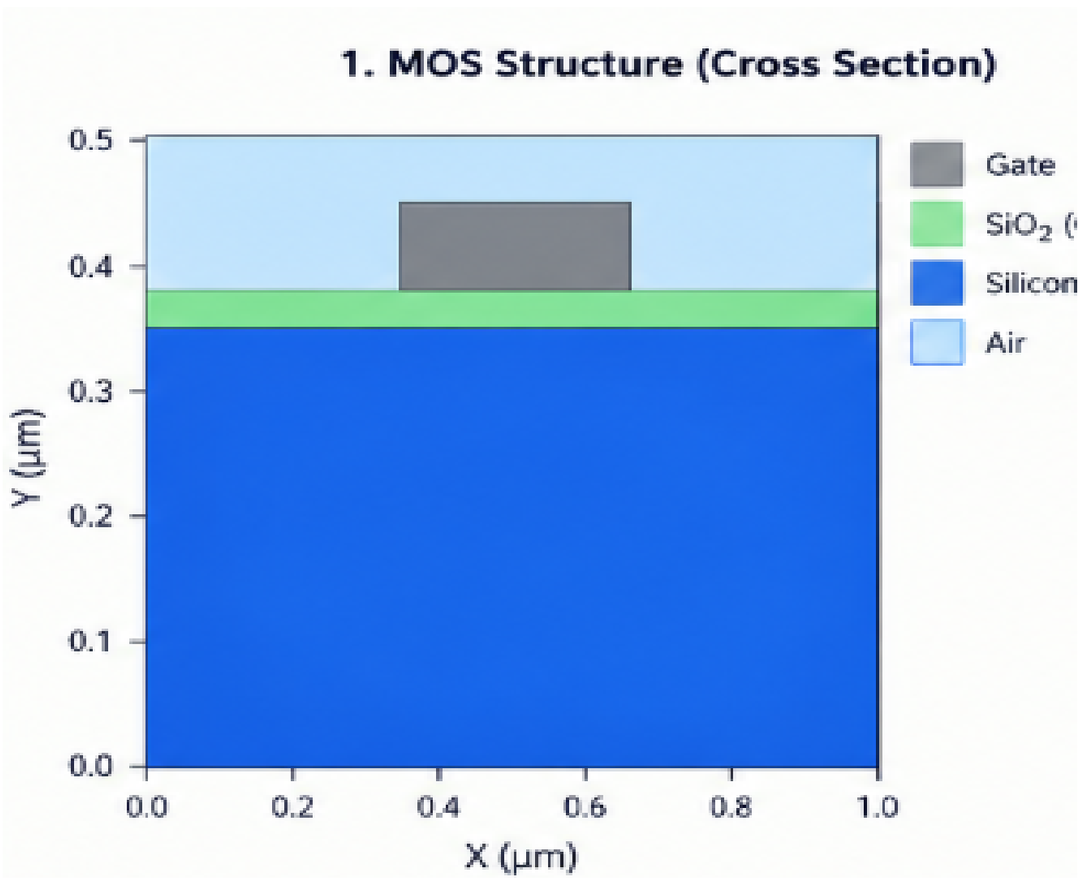
Circuit diagram:

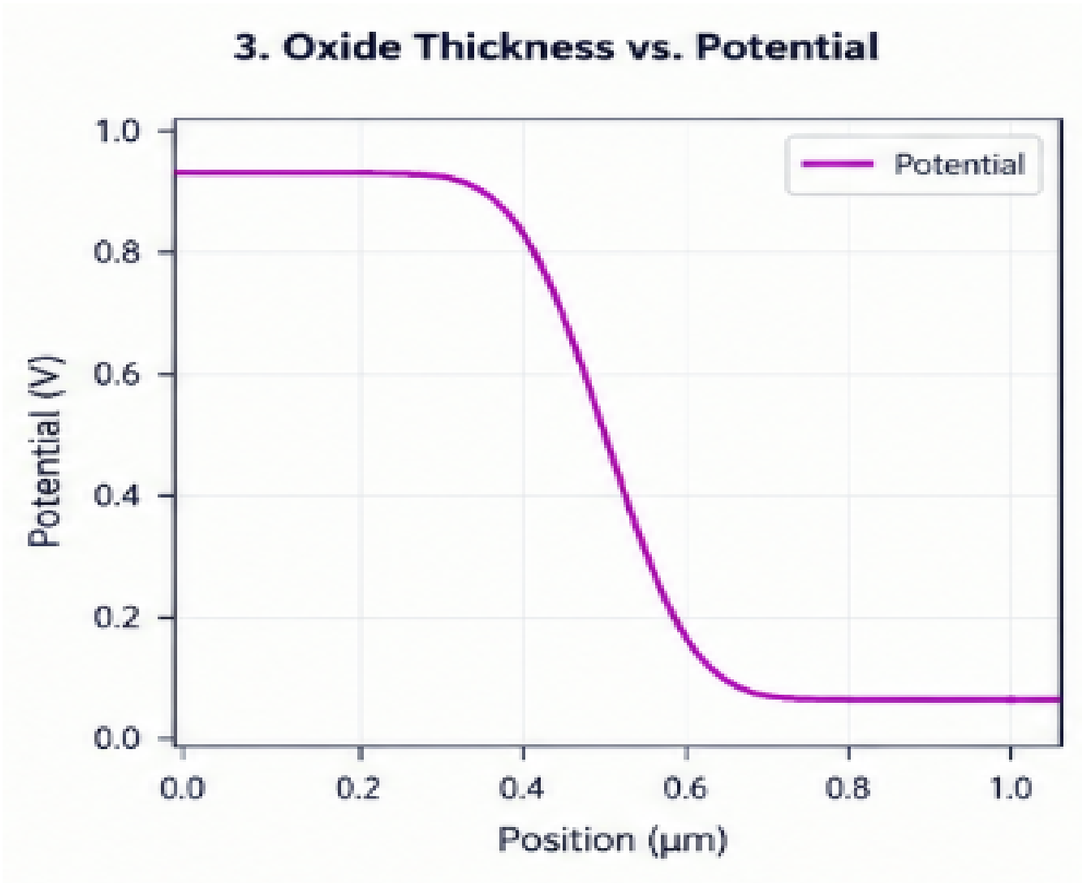


Output Parameter Table:

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
1	Oxidation process	Proper SiO_2 formation	SiO_2 layer formed	0% deviation
2	Gate oxide thickness	4 nm	4 nm	0% deviation
3	Gate length	180 nm	180 nm	0% deviation
4	Gate width	1 µm	1 µm	0% deviation
5	MOS layer structure	Si / SiO_2 / Gate	Si / SiO_2 / Gate	0% deviation
6	Oxide uniformity	Uniform layer	Uniform oxide layer observed	0% deviation
7	Gate capacitance	1.55 fF	1.55 fF	0% deviation
8	Interface/oxide quality	Proper MOS interface	Proper MOS interface observed	0% deviation

Output:


1. MOS Structure (Cross Section)
Gate
SiO_2 (
Silicon
Air
Y (µm)
X (µm)
0.0
0.1
0.2
0.3
0.4
0.5
0.0
0.2
0.4
0.6
0.8
1.0


3. Oxide Thickness vs. Potential
Potential
Potential (V)
Position (µm)
0.0
0.2
0.4
0.6
0.8
1.0